

Chapter 12: E-Commerce overview- Part II

Please make sure to read the textbook for details on topics below:

E-Commerce Security

- Encryption
 - Ensures privacy within an organization and on the Internet.
 - The conversion of data into an unreadable form, called a ciphertext.
- Decryption
 - The process of converting the ciphertext back into its original form, called plaintext or cleartext, so it can be understood.
- The encryption/decryption process requires an algorithm and a key.

E-Commerce Security Encryption Types

Secure E-Commerce transactions use the encryption technologies below:

- Symmetric-key Encryption
- Asymmetric-key Encryption
- Hash Encryption

SSL (Secure Sockets Layer)

- Utilizes these encryption technologies
- Provides for secure transmission of data on the Internet.

E-Commerce Security: Symmetric-Key

Symmetric-Key Encryption

- Also called single-key encryption
- Both encryption and decryption use the same key
- Both the sender and receiver must know the key before communicating using encryption.
- Advantage: speed

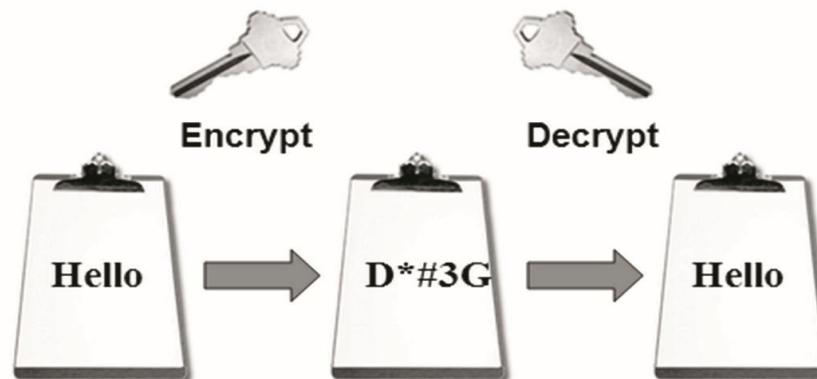


Figure 12.1 Symmetric-key encryption uses a single key

E-Commerce Security: Asymmetric-key

Asymmetric-Key Encryption

- Also called public-key encryption
- There is no shared secret
- Two keys are created at the same time:
 - Public key
 - Private key
- Asymmetric-key encryption is much slower than symmetric-key encryption.

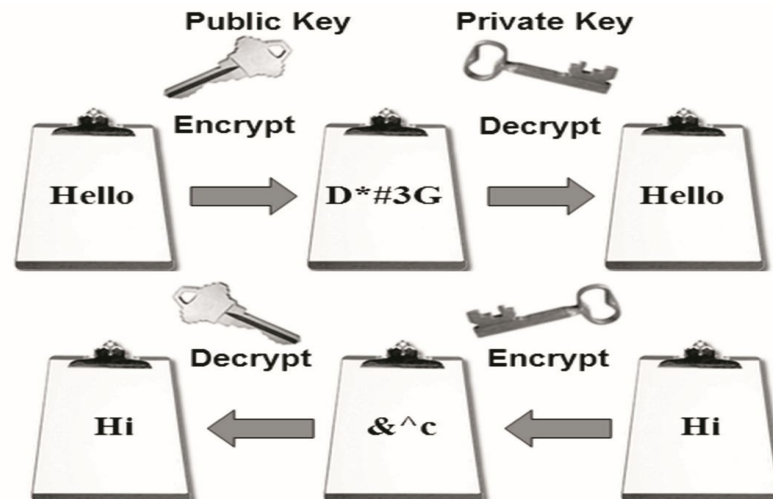


Figure 12.2 Asymmetric-key encryption uses a key pair

E-Commerce Security: Hash

Hash Encryption

A hash algorithm transforms a string of characters into a “digest”

- A shorter fixed-length value or key that represents the original string

One-way encryption

Used for information that will not be read or decrypted

Purpose: verify the integrity of information

Secure Sockets Layer (SSL)

- A protocol that allows data to be privately exchanged over public networks
- Encrypts data sent between a client (usually a Web browser) and a Web server.
- Utilizes both symmetric and asymmetric keys.
- “https” protocol
 - Browsers display a “lock” icon
 - Replaced by Transport Layer Security (TLS)

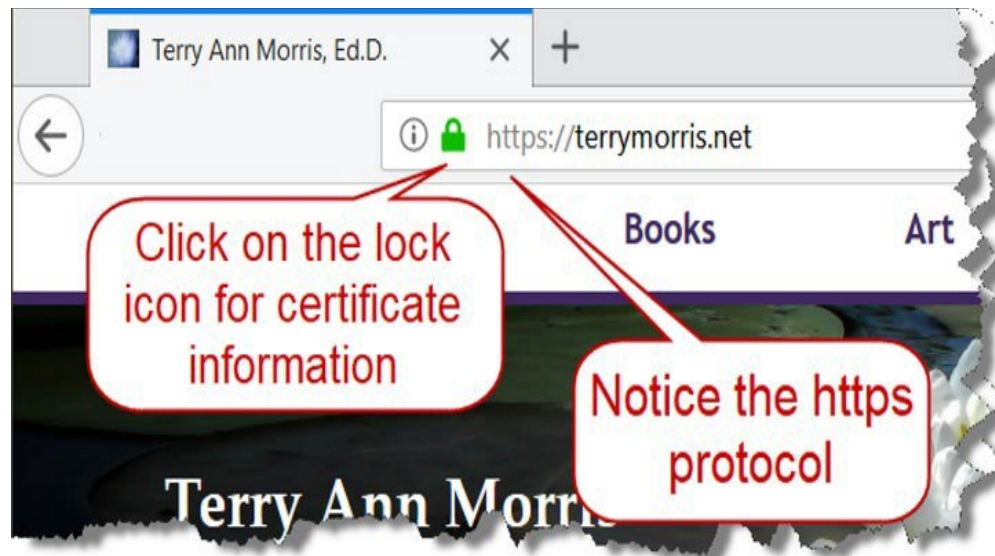


Figure 12.3 The browser indicates that SSL is being used. Screenshots of Mozilla Firefox. Courtesy of Mozilla Foundation.

SSL provides secure communication between a client and server by using:

- Server and (optionally) client digital certificates for authentication
- Symmetric-key cryptography using a "session key" for bulk encryption
- Public-key cryptography for transfer of the session key
- Message Digests (hash encryption) to verify the integrity of the transmission

SSL & Digital Certificate

Digital Certificate

- A form of an asymmetric key
 - Also contains information about the certificate, the holder of the certificate, and the issuer of the certificate.
- Used by SSL to authenticate the identity of the web server

Digital Certificate

The contents of a digital certificate include:

- The public key
- Effective date of the certificate
- Expiration date of the certificate
- Details about the Certificate Authority – the issuer of the certificate
- Details about the certificate holder
- A digest of the certificate content

Certificate Authority

A trusted third-party organization or company that issued digital certificates.

Well-known Certificate Authorities:

Verisign

<https://www.verisign.com>

Thawte

<https://www.thawte.com>

Entrust

<https://www.entrust.com>

Order & Payment Processing

E-Commerce Payment Methods:

- Credit Card
- Stored-value Card
- Smart Card
- Digital Wallet
- Digital Cash

E-Commerce Storefront Solutions

- Instant Online Storefront
 - Shopify, BigCommerce
- Shopping Cart Software
 - Agoracart, osCommerce, ZenCart
- Easy Entry to E-Commerce
 - PayPal
 - Shopping Cart & Payment Verification
- E-Commerce Platform
 - Microsoft Azure Commerce
 - Adobe Commerce Cloud

Summary

This chapter introduced you to e-commerce security concepts.